

Abhishek Tomar

Personal Information

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Work experience

Lead Machine Learning Engineer Observe.ai

📅 12/2024 – present 📍 BENGALURU, INDIA

Solving contact center problems using Generative AI.

- **InsightVertex** - Led the development of a scalable RAG architecture, integrating robust **vector search** with a hybrid of SoTA open-source & **commercial** LLMs for precise **semantic query processing** and deriving actionable insights from unstructured contact center data. Drove adoption across 20+ clients by architecting for **high-throughput, low-latency** model serving, enabling proactive customer problem resolution.

Senior Machine Learning Engineer Soroco

📅 11/2023 – 12/2024 📍 BENGALURU, INDIA

- **FlowGraphs** - Developed a key feature for team-based **knowledge representation** using **NLP** techniques by leveraging open-source **Large Language Models (LLMs)** like **Mistral** and **Llama3**. This involves fine-tuning these models using the **QLoRA (peft)** framework, leading to deployment across 50+ industry-leading companies and securing a million-dollar deal with a Fortune 500 company.
- **QueryHub** - Designed and implemented an advanced internal knowledge retrieval platform utilizing **Retrieval-Augmented Generation (RAG)**, aimed at optimizing the search and access process within the company's expansive wikis. This system uses open-source models like **Llama3** and **Mistral**, **ANNOY** for Approximate Nearest Neighbor (**ANN**) searches, and the **Bert** transformer for precise embeddings, enhancing productivity by providing quick access to relevant information.
- **AutoEval** - Designed and implemented an **evaluation pipeline** for benchmarking the performance of diverse open-source Large Language Models (LLMs) against tailored use-case scenarios. This pipeline, deployed using **docker**, significantly reduced manual labor by **hundreds of hours**.

Machine Learning Engineer Digital Aristotle (BYJUS Subsidiary)

📅 02/2021 – 10/2023 📍 BENGALURU, INDIA

- **Byjus QnA Search** - Developed a robust, **high-traffic** question search system handling over **2 million req/day**, integrating Similar Question and Auto Completion modules. Used **ElasticSearch**, **Transformer(Bert)**, **ANN** for effective question suggestions and Learning-To-Rank model for result ranking. This module helps in generating **~\$600k/month** in **revenue** through sales leads.
- **Universal Search Enhancement** - Developed universal search feature for Byjus app to fetch resources **beyond just questions**, utilizing the Bert Transformer for query intent recognition and **entity extraction**, and deployed using **Docker**. Achieved an **F1 score** of **0.96**, with Text **SSR** at **91%** and Image SSR around **60% in production**.
- **ByjuWizz** - Developed hyper-personalised products using **Large Language Models (LLMs)**, specifically **GPT4**, with **prompt engineering**, **Langchain**, and **Pinecone** for premium users. Products include Teacher GPT for problem-solving guidance, Math GPT for step-by-step math solutions, and a textbook-specific doubt solver.
- **ByjusLLM** - Evaluated and **fine-tuned** open-source Large Language Models (**LLM**) like **LLaMA**, **LLaMA2**, **Falcon** for applications in math reasoning and doubt-solving chatbots. The project involved fine-tuning LLMs using **QLoRA** and **Peft** framework. The LLaMA 13B model achieved an accuracy of 13% on GSM8k.
- **Finetuning Transformer** - Fine-tuned the Bert transformer on ed-Tech datasets using **siamese** and **highway network** architectures. Implemented **contrastive**, cosine, and cross-cross-entropy loss for similarity and classification tasks.
- **GAN based watermark removal** - Built a **cGAN**-based model for watermark removal from solution images, aiding Byjus dB data ingestion. Utilized OpenCV, U-Net with pre-trained **ResNet-50**, PatchCNN
- **Upper Body Detector** - Fine-tuned **YOLOv5** for upper body object detection on annotated classroom data. This model is used for classroom engagement. Achieved **mAP50-95** of **0.922** on YOLOv5l6.

Data scientist Axtria India Pvt Ltd

📅 06/2018 – 02/2021 📍 GURGAON, INDIA

- **HCP Call Planning** - Developed a call assignment system for Health Care Practitioners across 28 countries, using **unsupervised clustering** and call grids for new drugs. Also managed client engagement

Work experience

- and risk analysis.
- **Sales Forecasting** - Developed a model to forecast sales for a new drug, identifying the effect of promotions and key performance channels. Utilized **regression** for sales prediction and a **neural network** for channel impact analysis.
 - **Roster Integration Module Development** - Created a roster integration module for daily employee status tracking and event updates, using **Talend** for **data integration** and **PostgreSQL** for handling client input data quality and event creation.

Projects

- **Jigyasa.ai** - Developed a **search assistant** which is built on back of open source LLMs and search engine. Utilized **Ollama** for inference optimization.
- **MathLLM** - A novel LLM, based on **Mistral-7B**, SFT finetuned model, using **QLoRA**, on open-source and proprietary data. Improving Mistral accuracy on **GSM8K** by **+22%**.
- **Video Action Classification** - Implemented an **end-to-end trainable** two-stream fusion architecture for classifying videos based on the action performed. The project involved using **transfer learning** on **ResNet** and **VGG-16** and exploring fusion architectures for **spatial and temporal streams**.
- **Text Data Classification** - Developed a text classification system using **Support Vector Machine (SVM)**, **Principal Component Analysis (PCA)**, and **Auto-Encoders** for dimensionality reduction, with algorithms implemented in **PyTorch**.

Education

- IIT Delhi, India** 07/2016 – 06/2018
M.Tech in Computer Technology
GPA - 7.75/10
- BIET Jhansi, India** 07/2012 – 06/2016
B.Tech in Electronics & Communication Engineering
GPA - 8.06/10
- Class 12, CBSE, India** 05/2010 – 06/2012
PCM | GPA - 8.24/10
- Class 10, CBSE, India** 05/2009 – 05/2010
Science | GPA - 9.4/10

Skills

- **Programming** - Python | C++ | R | SQL | PostgreSQL | Matlab
- **Machine Learning Framework/Library** - PyTorch | Keras | Tensorflow | Numpy | Scipy | Pandas | ScikitLearn | OpenCV | PEFT (LoRA) | ONNX
- **Backend Development** - Flask | AWS EC2 | AWS S3 | MongoDB | Docker | Elasticsearch | Git | Grafana
- **Machine Learning Algorithm** - NLP | Deep Learning | Computer Vision | Natural Language Understanding | Explainable AI | Reinforcement Learning | Graph Neural Networks | Supervised Machine Learning | Unsupervised Machine Learning | Recurrent Neural Network | Convolutional Neural Network | Transformers (BERT) | LLM | FAISS | PineCone

Certificates

- Fundamentals of Scalable Data Science**
IBM
 10/2019
- Course covers big data analysis using scalable machine learning techniques and introduces Apache Spark for large-scale data processing
- Introduction to Deep Learning**
HSE University
 08/2022
- Course provides foundational knowledge on deep learning concepts, techniques, and their application in various fields
- Deep Generative Models**
IISC Bengaluru
 06/2023
- Course explores the mathematics and application of deep learning models designed to generate new data.
 - Grade : A+